

Name: _____

Roll Number: _____

Set B

Submission URL: <https://forms.gle/8PpxzTACkrATWkxG7>

Boilerplate Code: <https://shorturl.at/0AI3F>

Note : Helper code is provided. It is entirely your choice to use it or not.

Question 1: Inorder Predecessor in BST

Task. Given a binary search tree (BST) where each node contains a unique integer value, your task is to find the in-order predecessor of a given key K . The in-order predecessor of a node in a BST is the node with the largest value less than K . If no such node exists, output -1 .

Input. The BST is constructed by inserting N values into an initially empty tree. The first line of input contains the integer N , representing the number of values to insert. The second line contains N space-separated integers, which are the values to be inserted into the BST in the given order. The third line contains the integer D value of which the inorder predecessor is to be found.

Output. Your program should output a single integer: the value of the in-order predecessor of K , or -1 if no predecessor exists.

Example.

Input:

7

20 8 22 4 12 10 14

10

Output:

8

Question 2: Deleting Element from Max-Heap

You are given an array of integers (might or might not be sorted). Your first task is to implement a max-heap and then perform the `deleteMax()` operation to delete the largest element from the heap. Print the max-heap before and after deleting the element.

Input Format

- First line contains an integer n , the size of the heap
- Second line contains n space-separated integers representing a valid max-heap

Output Format

- Single line containing n space-separated integers representing the max-heap before deletion
- Single line containing $n - 1$ space-separated integers representing the max-heap after deletion

Sample Input

```
6
1 2 3 4 5 6
```

Sample Output

```
6 5 3 4 2 1    6 4 5 1 3 2
5 4 3 1 2      5 3 4 1 2
```